

```

12. Enter how many six-sided dice you want to roll:''')
13. numberOfDice = int(input('> '))
14.
15. # Set up a dictionary to store the results of each dice roll:
16. results = {}
17. for i in range(numberOfDice, (numberOfDice * 6) + 1):
18.     results[i] = 0
19.
20. # Simulate dice rolls:
21. print('Simulating 1,000,000 rolls of {} dice...'.format(numberOfDice))
22. lastPrintTime = time.time()
23. for i in range(1000000):
24.     if time.time() > lastPrintTime + 1:
25.         print('{}% done...'.format(round(i / 10000, 1)))
26.         lastPrintTime = time.time()
27.
28.     total = 0
29.     for j in range(numberOfDice):
30.         total = total + random.randint(1, 6)
31.     results[total] = results[total] + 1
32.
33. # Display results:
34. print('TOTAL - ROLLS - PERCENTAGE')
35. for i in range(numberOfDice, (numberOfDice * 6) + 1):
36.     roll = results[i]
37.     percentage = round(results[i] / 10000, 1)
38.     print(' {} - {} rolls - {}'.format(i, roll, percentage))

```

After entering the source code and running it a few times, try making experimental changes to it. On your own, you can also try to figure out how to do the following:

- Try rolling 8-, 10-, 12-, or 20-sided dice.
- Try simulating two-sided coin tosses.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `lastPrintTime + 1` on line 24 to `lastPrintTime + 2`?
2. What bug do you cause if you delete or comment out `results[total] = results[total] + 1` on line 31?
3. What error happens if the user types letters instead of a number for the number of six-sided dice to roll?